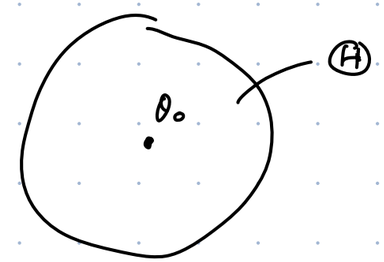
#10.

Eg. 10.5. $X_1 \dots X_n \overset{iid}{\sim} N(\theta, 1)$

$H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$



$$\hat{\theta}_{MLE} = \arg\max_{\theta \in \Theta} L(\theta | \mathbf{x})$$

LRT statistic

$$\lambda(\mathbf{x}) = \frac{\sup_{\theta=\theta_0} L(\theta|\mathbf{x})}{\sup_{\theta\in\Theta} L(\theta|\mathbf{x})} = \frac{L(\theta_0|\mathbf{x})}{L(\hat{\theta}_{MLE}|\mathbf{x})}$$

$$L(\theta|\mathbf{x}) = f(x_1 \dots x_n|\theta) = \prod_{i=1}^n f(x_i|\theta) = (2\pi)^{-\frac{n}{2}} \exp\left\{-\sum_{i=1}^n \frac{(x_i-\theta)^2}{2}\right\}$$

$$= \frac{(2\pi)^{-\frac{n}{2}} \exp\left\{-\sum_i \frac{(x_i-\theta_0)^2}{2}\right\}}{(2\pi)^{-\frac{n}{2}} \exp\left\{-\sum_i \frac{(x_i-\hat{\theta})^2}{2}\right\}} \qquad \hat{\theta} = \bar{x}$$

$$= \exp\left\{\left(-\underbrace{\sum_{i=1}^n (x_i-\theta_0)^2}_{= \sum_{i=1}^n (x_i-\bar{x})^2 + n(\bar{x}-\theta_0)^2} + \sum_{i=1}^n (x_i-\bar{x})^2\right)/2\right\}$$

$$= \exp\left\{-n(\bar{x}-\theta_0)^2/2\right\}$$

$$R = \{\mathbf{x} : \lambda(\mathbf{x}) \le c\} = \{\mathbf{x} : \exp\{-n(\bar{x}-\theta_0)^2/2\} \le c\} = \{\mathbf{x} : |\bar{x}-\theta_0| \ge \sqrt{-2\log(c)/n}\}$$

Eg. 10.7

$X_1 \cdots X_n \overset{iid}{\sim} N(\theta, \sigma^2)$ ← known

$H_0: \theta \le \theta_0$ vs $H_1: \theta > \theta_0$

$$R = \left\{ \mathbf{x} : \frac{\bar{x} - \theta_0}{\sigma/\sqrt{n}} > c \right\} \rightarrow \text{課題}$$

$$\beta(\theta) = P_\theta(\mathbf{X} \in R)$$

$$= P_\theta\left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > c \right)$$

$$= P_\theta\left(\frac{\bar{X} - \theta}{\sigma/\sqrt{n}} + \frac{\theta - \theta_0}{\sigma/\sqrt{n}} > c \right)$$

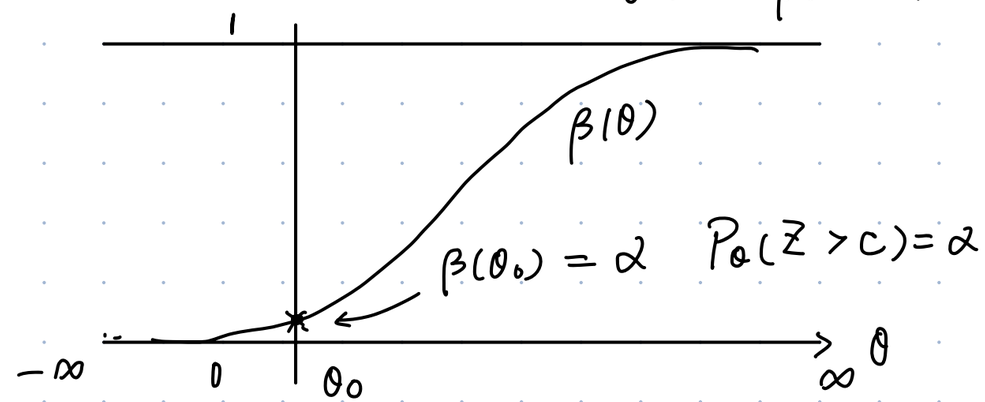
$$= P_\theta\left(Z > c - \frac{\theta - \theta_0}{\sigma/\sqrt{n}} \right)$$

$$= P_\theta\left(Z > c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}} \right)$$

$$= 1 - \Phi\left(c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}} \right)$$

$\Rightarrow$

$\theta \to \infty \quad \beta(\theta) \to 1$



$\theta \to -\infty$

$\beta(\theta) \to 0$

Eg 10.9. (Eg. 10.5)

$$\lambda(\mathbf{x}) = \exp\{-n(\bar{x} - \theta_0)^2/2\}$$

$$R = \{\mathbf{x} : \lambda(\mathbf{x}) \le c\}$$

$$\alpha = \sup_{\theta = \theta_0} \beta(\theta) = \sup_{\theta = \theta_0} P_\theta(\lambda(\mathbf{X}) \le c)$$

$$= P_\theta\left(-\frac{n(\bar{x} - \theta_0)^2}{2} \le \log c \right)$$

$$= P_\theta\left\{ \frac{n(\bar{x} - \theta_0)^2}{n} \ge -(2\log c) \right\}$$

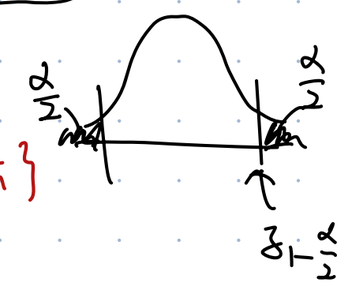
$$= P_\theta\left\{ \frac{\sqrt{n}|\bar{x} - \theta_0|}{\sqrt{n}} \ge \sqrt{-2\log c} \right\}$$

$$\sqrt{n}(\bar{X} - \theta_0) = \frac{\bar{X} - \theta_0}{\frac{1}{\sqrt{n}}} \sim N(0,1)$$

$$\| \; Z$$

$$= P_\theta\{|Z| \ge z_{1-\frac{\alpha}{2}}\}$$

$$= P_\theta\{|\bar{x} - \theta_0| \ge z_{1-\frac{\alpha}{2}}/\sqrt{n}\}$$



Thm 10.15 $\quad H_0: \theta = \theta_0 \quad$ vs. $\quad H_1: \theta \neq \theta_0$

$\log L(\theta|\mathbf{x}) = \ell(\theta|\mathbf{x})$.

$$= \ell(\hat\theta|\mathbf{x}) + \ell'(\hat\theta|\mathbf{x})(\theta-\hat\theta) + \ell''(\hat\theta|\mathbf{x})\frac{(\theta-\hat\theta)^2}{2!} + \cdots$$ Taylor expansion around $\hat\theta$

$$\ell(\theta_0|\mathbf{x}) = \ell(\hat\theta|\mathbf{x}) + \ell'(\hat\theta|\mathbf{x})(\theta_0-\hat\theta) + \ell''(\hat\theta|\mathbf{x})\frac{(\theta_0-\hat\theta)^2}{2!} + \cdots$$ ($\theta = \theta_0$ 代入)

$$\ell(\theta_0|\mathbf{x}) - \ell(\hat\theta|\mathbf{x}) \approx \underbrace{\ell'(\hat\theta|\mathbf{x})(\theta_0-\hat\theta)}_{=0} + \ell''(\hat\theta|\mathbf{x})\frac{(\theta_0-\hat\theta)^2}{2!}$$

$\frac{\partial}{\partial\theta}\ell(\theta|\mathbf{x}) = 0$

$\Rightarrow \theta = \hat\theta$ (MLE) $\Leftrightarrow \ell'(\hat\theta|\mathbf{x}) = 0$

$$\ell(\theta_0|\mathbf{x}) - \ell(\hat\theta|\mathbf{x}) \approx \frac{\ell''(\hat\theta|\mathbf{x})(\theta_0-\theta)^2}{2!}$$

Thm 9.37.

$$\sqrt{n}(\theta_0-\hat\theta) \xrightarrow{D} N\left(0, \frac{1}{I(\theta_0)}\right)$$

$$-2\log\lambda(\mathbf{x}) = -2\log\left(L(\theta_0|\mathbf{x}) - L(\hat\theta|\mathbf{x})\right)$$

$$= -2\left(\ell(\theta_0|\mathbf{x}) - 2\ell(\hat\theta|\mathbf{x})\right) \approx \underbrace{-\ell''(\hat\theta|\mathbf{x})}\,\underbrace{(\theta_0-\hat\theta)^2} \approx \{\sqrt{n}(\theta_0-\hat\theta)\}^2 I(\theta_0) = Z^2 \xrightarrow{D} (N(0,1))^2$$

χ^2_1

$$-\frac{1}{n}\ell''(\hat\theta|\mathbf{x}) \xrightarrow{P} I(\theta_0)$$ Report 11-2(b)

Fisher information